

1.9 Pre-Quiz: Arithmetic and Geometric Sequences

Review for Wednesday's quiz. A calculator is allowed. Show your working: write the formula you use, then substitute.

1. Consider the arithmetic sequence

$$12, 19, 26, 33, \dots$$

- (a) Write down the value of the common difference d .
- (b) Find an expression for u_n , the n^{th} term.
- (c) Find u_{40} .
- (d) Show that 250 is a term of the sequence, and find which term it is.
- (e) Determine whether 300 is a term of the sequence. Justify your answer.

(a) $19 - 12 = 7$

(b) $u_n = 12 + 7(n-1)$

(c) $u_{40} = 12 + 7(40-1)$
 $= 285$

(d) $u_n = 12 + 7(n-1) = 250$

$$n-1 = (250-12)/7 = 34$$

$$n = 35 \text{ since } 35 \in \mathbb{N}$$

$u_{35} = 250$ is in the sequence

(e) $u_n = 12 + 7(n-1) = 300$

$$n = (300-12)/7 + 1 = 42.14\dots$$

$42.14\dots \notin \mathbb{N}$ so 300 is NOT

in the sequence (also $u_{42} = 298$
 $u_{43} = 306$)

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2. In an arithmetic sequence, $u_5 = 31$ and $u_{12} = 3$.

(a) Find the common difference d and the first term u_1 .

(b) Find an expression for u_n .

(c) Find the first term of the sequence that is negative.

The first three terms of a different arithmetic sequence are

$$k + 2, \quad 3k, \quad 4k + 1.$$

(d) Find the value of k , and write down the first three terms.

(e) Find the 20th term of this sequence.

$$\begin{aligned} (a) \quad u_{12} - u_5 &= [u_1 + d(12-1)] - [u_1 + d(5-1)] = \cancel{31} - 31 \\ &7d = -28 \\ &d = -4 \end{aligned}$$

$$\begin{aligned} (b) \quad u_5 &= u_1 + -4(5-1) = 31 \\ u_1 &= 31 + 16 = \cancel{47} \\ u_n &= 47 - 4(n-1) \end{aligned}$$

$$\begin{aligned} (c) \quad \cancel{u_n} \quad u_n &= 47 - 4(n-1) < 0 \\ -4n &< \cancel{47} - 51 \\ n &> 12.75 \text{ so } n = 13 \end{aligned}$$

$$u_{13} = 47 - 4(13-1) = -1$$

$$\begin{aligned} (d) \quad 3k - (k+2) &= 4k+1 - 3k \\ 2k-2 &= k+1 \end{aligned}$$

$$k = 3$$

$$5, 9, 13 \quad d=4$$

$$(e) \quad u_{20} = 5 + \frac{4}{1}(20-1) = 81$$

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3. (a) State whether each sequence is arithmetic, geometric or neither. Write down d or r where there is one.

(i) 3, 7, 11, 15 (ii) 3, 6, 12, 24 (iii) 3, 4, 6, 9 (iv) 80, 20, 5, 1.25

Consider the geometric sequence

6, 9, 13.5, ...

- (b) Find the common ratio r .
(c) Write down an expression for u_n .
(d) Find u_{10} .
(e) Find the smallest value of n for which $u_n > 1000$.
(f) A different geometric sequence has $u_2 = 12$ and $u_5 = 324$. Find the common ratio and the first term.

i) arithmetic $d=4$

ii) geometric $r=2$

iii) neither

iv) geometric $r = \frac{1}{4}$

(b) $r = \frac{9}{6} = 1.5$

(c) $u_n = 6 \cdot (1.5)^{n-1}$

(d) $u_{10} = 6 \cdot 1.5^{10-1} = 230.6 \approx 231$

(e) $u_n = 6 \cdot (1.5)^{n-1} > 1000$

$u_{13} = 778.5$, $u_{14} = 1167.75$... so $n=14$

(f) $\frac{u_5}{u_2} = \frac{u_1 \cdot r^{5-1}}{u_1 \cdot r^{2-1}} = \frac{324}{12}$
 $r^3 = 27$ $r=3$

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4. Elena is offered two jobs.

Job A: \$42 000 in the first year, rising by \$1500 each year.

Job B: \$40 000 in the first year, rising by 5% each year.

- (a) Find Elena's salary in year 3 for each job.
- (b) Write down an expression for the salary in year n for each job.
- (c) Find the first year in which the Job B salary is greater than the Job A salary.

Elena invests \$3000 at 3% per year compounded annually.

- (d) Find the value of the investment after 8 years.
- (e) Inflation is 2% per year. Find the real value of the investment after 8 years.

4a) A: $u_3 = 42,000 + 1500(3-1) = 45,000$
B: $u_3 = 40,000 \times (1.05)^{3-1} = 44,100$

b) A: $u_n = 42,000 + 1500(n-1)$
B: $u_n = 40,000 \times (1.05)^{n-1}$

(c)

n	A	B	
4	46,500	46,305	year 5
5	48,000	48,620.25	

d) $FV = 3000(1.03)^8 = \$3800.31$

e) Real Value = $\frac{3800.31}{1.02^8} = \3243.53